William Fernando CHAPARRO-PICO

Microbiologists | MSc in Microbiology

Web page | ORCID: 0000-0002-9028-710X | LinkedIn

Paris, France

PROFILE

Microbiologist with a Master's degree in Microbiology and a research background in respiratory virology and infectious diseases. Experienced in molecular and cellular biology, respiratory viral diagnostics, tropical infectious disease research and diagnostics, laboratory workflow implementation, sequencing technologies, and bioinformatic and statistical analysis of molecular and epidemiological data. Skilled in respiratory clinical sample collection and processing, RNA/DNA extraction, cell culture, multiple PCR-based approaches, and protocol optimization, with hands-on experience working under BSL-1 and BSL-2 conditions and applying BSL-3 laboratory practices. Background in molecular virology, immunology, and host–pathogen interactions, complemented by teaching experience in molecular biology and immunology for undergraduate biology students. Interested in research integrating respiratory viruses, molecular diagnostics, host–pathogen biology, and translational biomedical research.

EDUCATION

Master's degree in Microbiology

Universidad Industrial de Santander, School of Microbiology, Bucaramanga, Colombia

2023 – 2025

- **Thesis:** Molecular epidemiology of Respiratory Syncytial Virus and Rhinovirus in the Santander population, Colombia, 2020–2024.

Bachelor's degree in Microbiology and Bioanalysis

Universidad Industrial de Santander, School of Microbiology, Bucaramanga, Colombia

2017 – 2022

- **Thesis:** Evaluation of physical, chemical, and physicochemical pretreatments of saliva samples for SARS-CoV-2 diagnosis by direct RT-qPCR.

PROFESSIONAL EXPERIENCE

Research Engineer in Microbiology and Molecular Biology (DNA Sequencing)

PSL Proanálisis S.A.S. – Ecopetrol, Piedecuesta, Colombia

July 2025 – April 2026

- **Biohydrogen production potential**: Design and implementation of metagenomic assays to characterize the microbial biodiversity associated with oilfield environments with potential for biohydrogen production. I developed and standardized DNA sequencing workflows—from sample collection to data analysis—to identify potential H_2-producing bacteria.

- **Microbial diversity in oil-contaminated environments**: Performed molecular and bioinformatic analyses of soil and water samples from oil-contaminated environments to characterize microbial communities.

Research Technician in Chagas disease cardiomyopathy: biomarkers of cardiac tissue

Universidad Industrial de Santander – Johns Hopkins University – Fundación Cardiovascular de Colombia

2024 — 2025

- Processed cardiac tissue explants from heart transplant patients, performing cardiac dissection and tissue sampling for downstream isolation and analysis of RNA, proteins, and inflammatory infiltrate cells.

Research Technician in Viral Diagnostics

Universidad Industrial de Santander, School of Microbiology, Bucaramanga, Colombia

March 2023 – May 2023

- Performed viral sample collection, DNA/RNA extraction, assay standardization, and RT-qPCR detection of Respiratory Syncytial Virus and Rhinovirus in respiratory tract samples from patients with acute respiratory infection.

Research Technician

Universidad Industrial de Santander, School of Microbiology, Bucaramanga, Colombia

August 2022 – December 2022

- Performed molecular biology and immunology based assays for infectious disease diagnosis, with emphasis on *Trypanosoma cruzi*.

- Report writing, administrative management, and database updates for the Immunology and Molecular Epidemiology Research Group (GIEM-UIS).

TEACHING EXPERIENCE

Part-time Lecturer in General Biology
Universidad Industrial de Santander, School of Microbiology, Bucaramanga, Colombia
February 2025 – June 2025

- Taught general biology through lectures and practical laboratory sessions for first-semester undergraduate students in Microbiology and Bioanalysis, with emphasis on fundamental biological concepts, laboratory techniques, and scientific practice.

Part-time Lecturer in Immunology
Universidad Industrial de Santander, School of Microbiology, Bucaramanga, Colombia
February 2025 – June 2025; February 2024 – June 2024

- Taught immunology through lectures and practical laboratory sessions, with emphasis on immunological techniques and their applications in clinical diagnostics and biomedical research. Supervised undergraduate research seminars focused on immune responses to bacterial, viral, and fungal pathogens.

Part-time Lecturer in Molecular Biology
Universidad Industrial de Santander, School of Microbiology, Bucaramanga, Colombia
August 2024 – December 2024; January 2023 – February 2023

- Taught molecular biology through lectures and practical laboratory sessions for undergraduate students in Microbiology and Bioanalysis, with emphasis on molecular techniques, experimental approaches, and core laboratory practices.

PUBLICATIONS

Chaparro-Pico, W. F., Bueno, N., Lozano-Parra, A., Niederbacher, J., Herrera, V., Sosa Ávila, L. M., Machuca Pérez, M. A., & Díaz Galvis, M. L. (2026). **Molecular Epidemiology of Respiratory Syncytial Virus and Rhinovirus in Santander, Colombia, During the COVID-19 Pandemic and Post-Pandemic Periods, 2020–2024**. Viruses, 18(6), 666. doi: 10.3390/v18060666

Carbone-Schellman, J., Fontecilla-Escobar, J., Sales-Salinas, N., **Chaparro-Pico, W. F.**, Molina-Berríos, A., Ruete, M. C., González, P. A., Machuca, M. A., Opazo, M. C., Ezquer, M. E., and Duarte, L. F. (2025). **Therapeutic challenges in central nervous system viral infections: Advancing mesenchymal stem cell-based strategies for treating neuroinflammation and promoting tissue repair**. *Frontiers in Immunology*, 16. doi: 10.3389/fimmu.2025.1677433.

Sosa, L. M., Díaz, M. L., Lozano-Parra, A., Bueno, N., Niño, D., Herrera, V., Machuca, M. A., **Chaparro-Pico, W. F.**, & Niederbacher, J. (2025). **Virus respiratorios en pacientes con síntomas de SARS-CoV-2 durante el periodo pandémico en Santander, Colombia**. *Salud UIS*, 57, e25v57a19. doi: 10.18273/saluduis.57.e:25v57a19.

SCIENTIFIC PRESENTATIONS

Oral presentations

Molecular epidemiology of Rhinovirus infection: A66 genotype and fatal outcome in a case of acute respiratory infection.
4th International Health Science Meeting: Science Transforms, Colombia.
September 2024

Validation of a rapid immunochromatographic assay for the diagnosis of Chagas disease in the Santander population.
U23 Fest International Congress: United for our well-being, Colombia.
November 2023

Single nucleotide variant of the *ITGAM* gene is associated with the development of Chronic Chagasic Cardiomyopathy.
VIII Colombian Meeting on Leishmaniasis and Chagas Disease, Colombia.

August 2023

Evaluation of physical, chemical, and physicochemical treatments for SARS-CoV-2 diagnosis by direct RT-qPCR.
XVIII Congress of the Colombian Association of Parasitology and Tropical Medicine, Colombia.
December 2022

Seroprevalence of *Trypanosoma cruzi* infection and characterization of patients with chronic Chagasic cardiomyopathy from five municipalities in Colombia.
III International Meeting of Health Sciences and I International Congress on Life, Health and Well-being, Colombia.
November 2022

Identification of genetic markers associated with Chagas disease in Latin American population.
U21 International Congress – Sustainable Development Goals, Colombia.
November 2021

Posters

Molecular characterization of Respiratory Syncytial Virus and Rhinovirus in respiratory samples from Santander, Colombia, 2020–2024.
XXVI Latin American Congress of Clinical Biochemistry and XXII International Congress of the National College of Bacteriology, Colombia.
October 2024

Differential circulation of Rhinovirus species and genotypes/clades during the COVID-19 pandemic and post-pandemic period.
XIV National Meeting on Research and Innovation in Infectious Diseases and IV Latin American Meeting on Research in Infectious Diseases, Colombia.
September 2024

Detection of respiratory viruses during the introduction and first wave of SARS-CoV-2.
XX Latin American Congress of Pediatric Infectious Diseases.
November 2023

Detection of viruses causing acute respiratory infection during the SARS-CoV-2 pandemic period, 2020–2021.
U23 Fest International Congress: United for our well-being, Colombia.
November 2023

Chagas disease in a population assisted in an elderly care center in Bucaramanga.
U23 Fest International Congress: United for our well-being, Colombia.
November 2023

Replication study of genetic markers associated with Chagas disease in Latin American population.
IX Symposium on Molecular Biology of Chagas Disease, Colombia.
June 2022

COMPLEMENTARY ACADEMIC TRAINING

Courses

Portable sequencing techniques (MinION) applied to environmental monitoring
Alexander von Humboldt Institute and Ecopetrol, Colombia
2025

Theoretical and Practical Introductory Course on Flow Cytometry and Operation of the Attune NxT Instrument
Thermo Fisher Scientific, Colombia
2023

Pathobiology of Intracellular Infections
Yale University; Fogarty International Center; International Center for Medical Training and Research, CIDEIM,

Colombia
2023

IX International Course on Trypanosomatids
Universidad de Antioquia, Colombia
2022

High-content and high-predictive cellular models for host–pathogen interaction studies, disease modeling, and drug discovery
Institut Pasteur, Montevideo, Uruguay
2022

TECHNICAL SKILLS

Molecular Biology: DNA/RNA extraction and purification; RT-qPCR; RT-PCR; conventional PCR; restriction enzyme digestion; Western blotting; protein electrophoresis; agarose gel electrophoresis.

Sequencing, Genomics, and Bioinformatics: Oxford Nanopore Technologies (ONT) sequencing; genome and metagenomic analysis; phylogenetic and evolutionary analysis (ML via IQ-Tree and Bayesian via Beast 2); Sanger sequencing; Multiple sequence alignment; DNA sequence quality control and preprocessing; primer design.

Microbiology, Virology, Immunology, and Cell-Based Methods: BSL-1 and BSL-2 laboratory practices applyinf BSL/3 laboratory practices; handling of viral pathogens; respiratory viral sample collection and processing; mammalian cell culture; bacterial isolation and culture; flow cytometry; ELISA; immunological assays.

Data Analysis and Scientific Computing: Python (basic); R (basic); Linux (basic); LaTeX; Microsoft Office.

Languages: English, advanced (B2-C1); French, Beginner; Spanish, native

GRANTS AND FELLOWSHIPS

Master's Scholarship
High-level human capital development for the Department of Santander at the Universidad Industrial de Santander, BPIN 2020000100536.